Group Coursework Submission Form

Specialist Masters Programme

Please list all names of group members: (Surname, first name) 1. Devasagaya, Christy 2. Mojumder, Abdus Sattar 3. Norena Jaramillo, Luis Fernando	4. Pinto, Aaron Neil **GROUP NUMBER:** 7
MSc in: Actuarial Science	
Module Code: SMM048	
Module Title: Insurance Risk Modelling (Subject CS2)	
Lecturer: Dr. Michail Chronopoulos	**Submission Date: 21 March 2025**

Declaration:

By submitting this work, we declare that this work is entirely our own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the coursework instructions and any other relevant programme and module documentation. In submitting this work we acknowledge that we have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. We also acknowledge that this work will be subject to a variety of checks for academic misconduct.

We acknowledge that work submitted late without a granted extension will be subject to penalties, as outlined in the Programme Handbook. Penalties will be applied for a maximum of five days lateness, after which a mark of zero will be awarded.

Marker's Comments (if not being marked on-line):

Deduction for Late Submission of assignment:

For Students:
Once marked please refer to Moodle for your final coursework grade, including your Peer Assessment grade.

SMM048: Insurance Risk Model
Coursework 2

Submitted to

Dr. Michail Chronopoulos

Submitted by

Christy Devasagaya
Abdus Sattar Mojumder
Luis Fernando Norena Jaramillo
Aaron Neil Pinto

Date: March 21, 2025

Bayes Business School
City St George's, University of London
Masters in Actuarial Science

Contents

1 Question 1 **2**
- 1.1 Normality and Independence of Daily Increments . . . 2
- 1.2 Effect of Log Transformation . . . 5
- 1.3 Discretisation of raw Data . . . 5
- 1.4 Markov Chain Fitting and Likelihood Ratio Test . . . 8
- 1.5 Confidence Interval Construction . . . 9
- 1.6 Factors Influencing the Fluctuations . . . 10

2 Question 2 **12**
- 2.1 Types of Censoring in Data . . . 12
- 2.2 Kaplan-Meier Estimate of the Time in Remission . . . 12
  - 2.2.1 Construct a Kaplan-Meier estimate . . . 12
  - 2.2.2 Graphical Representation of Survival and Hazard Function . . . 15
  - 2.2.3 Conclusion on Whether the New Treatment Results in Greater Remission Time . . . 16
- 2.3 Kaplan-Meier Assumptions and Applicability . . . 17
- 2.4 Probability of Remission After 30 Weeks . . . 17
- 2.5 Hypothesis Test for Remission Probability . . . 18

3 Question 3 **20**
- 3.1 Probability of Functioning After 60 Hours . . . 20
- 3.2 Conditional Survival Probability . . . 21
- 3.3 Median Failure Time Calculation . . . 21
- 3.4 Alternative Hazard Function Formulation . . . 22
- 3.5 Baseline Hazard Function . . . 22
- 3.6 Survival Function Comparison . . . 23
- 3.7 Impact of Driving Style on Model Accuracy . . . 24
- 3.8 Additional Covariates for Model Improvement . . . 25

1 Question 1

This analysis explores the modelling of financial time series data using two distinct approaches: a random walk model with normally distributed increments and a random walk model with discrete increments. The study utilises daily closing prices of the **Nikkei 225**and **FTSE 100** indices, sourced from Yahoo Finance for the period **1 August 2016 to 31 July 2017**.

1.1 Normality and Independence of Daily Increments

The primary objective is to examine the statistical properties of daily price increments, assess the effectiveness of log transformation, and determine whether a discretised Markov Chain model provides a superior fit compared to a traditional independent increment model. By systematically evaluating these aspects, this study aims to identify dependencies in market movements and establish a more effective modelling framework for financial forecasting.

Structured Progression of the Analysis

1. **Normality Test**
   We have a sequence of observations $x_1, \ldots, x_n$ representing the closing prices of the index. To analyse whether the daily increments are normally distributed, a standard method is to investigate whether the increment depends on x_n, which represents the closing price of the index on day n:

   $$j_{n+1} = x_{n+1} - x_n$$

   (**Note:** The Nikkei 225 dataset contains **missing values** due to non-functioning days of the stock market. Therefore, we have **cleaned the data** as required for accurate analysis.)

   The histograms in figure 1 provide a visual representation of the distribution of daily price increments for both indices. Additionally, histograms of the log-transformed data were generated to assess whether the transformation improves normality.

   The histograms of raw daily increments exhibit a bell-shaped central region, aligning with a normal distribution. However, fat tailsare evident, indicating a higher occurrence of extreme price movements than expected under a perfect normal distribution. This suggests that while price changes approximate normality, they exhibit heavy-tailed behaviour, deviating from strict Gaussian assumptions.

   To further assess normality, we perform the **Shapiro-Wilk test**, a standard statistical test for detecting deviations from normality. The results for the test are tabulated in table 1 below.

Result Interpretation:
Since all p-values (shown in the table 1) are below 0.05, we reject the null hypothesis of normality. This confirms that daily price increments are **not normally distributed**, reinforcing the observation of **fat-tailed behaviour** seen in the histograms

Dataset	W Statistic	p-value	Normality Conclusion
N225_dif_numeric	0.91656	1.877×10^{-10}	Not normal
N225_dif_numeric_log	0.89407	4.599×10^{-12}	Not normal
FTSE_dif_numeric	0.98288	0.004088	Not normal
FTSE_dif_numeric_log	0.98120	0.002105	Not normal

Table 1: Shapiro-Wilk normality test results for daily price increments and log-transformed data.

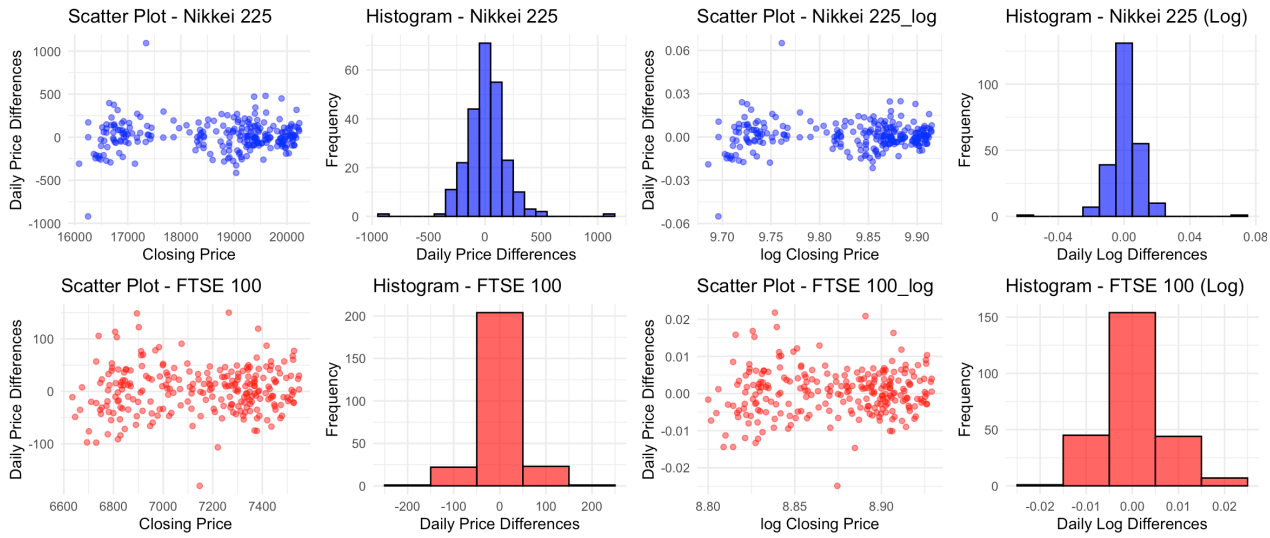


Figure 1: Scatter plots and histograms of daily price differences and log-transformed differences for the Nikkei 225 and FTSE 100 indices.

2. **Independence test**

   A scatter plot (shown in figure 1) of daily increments against previous closing prices was examined to assess the presence of **heteroskedasticity**, i.e., variance instability. The scatter plot shows a stable variance, with no cone-shaped pattern, indicating that variance stabilisation through log transformation is unnecessary. The random spread of points supports the assumption that daily increments are **independent** and do not systematically depend on past closing prices.

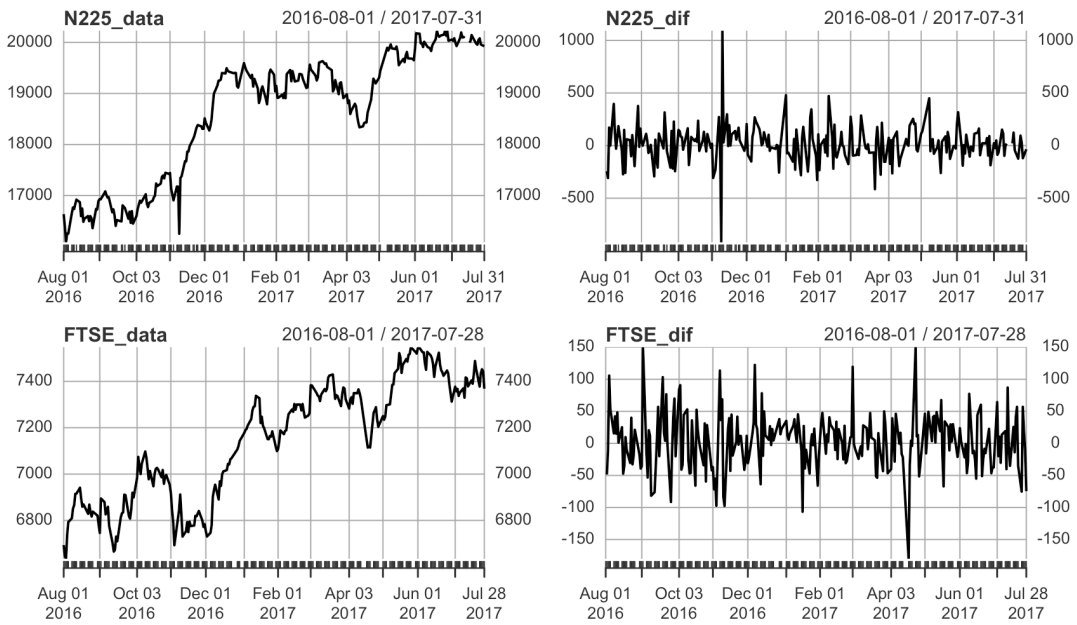


Figure 2: Graphs of the closing prices and price increments for the Nikkei 225 and FTSE 100 indices. The top-left plot shows the closing prices for the Nikkei 225, the top-right plot shows the price increments for the Nikkei 225, the bottom-left plot shows the closing prices for the FTSE 100, and the bottom-right plot shows the price increments for the FTSE 100.

1.2 Effect of Log Transformation

To further assess independence, we plotted the **Autocorrelation Function (ACF)** for daily price increments. The ACF plot shows that all autocorrelation values lie within the blue confidence bands, indicating that no significant correlation exists between successive increments. This confirms that daily price movements do not exhibit strong autocorrelation, consistent with the random walk hypothesis.

Result Interpretation:
Since variance remains stable and autocorrelation is negligible, we conclude that **daily price increments are independent**, aligning with the assumptions of a random walk model. Log transformation does not improve normality significantly, as evidenced by the histograms and statistical tests. Therefore, we proceed with the original increments without transformation for further analysis.

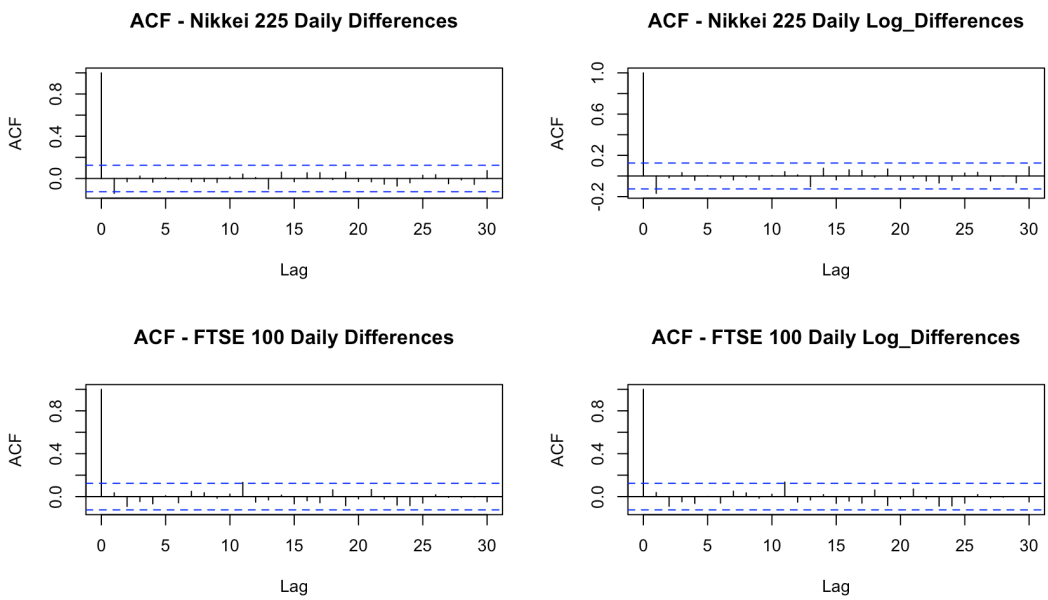


Figure 3: Autocorrelation function (ACF) plots for the daily differences and daily log differences of the Nikkei 225 and FTSE 100 indices. The top-left plot shows the ACF for the Nikkei 225 daily differences, the top-right plot shows the ACF for the Nikkei 225 daily log differences, the bottom-left plot shows the ACF for the FTSE 100 daily differences, and the bottom-right plot shows the ACF for the FTSE 100 daily log differences.

The daily price increments of the Nikkei 225 and FTSE 100 indices do not follow a normal distribution due to heavy-tailed behaviour, but they exhibit independence over time, supporting the random walk hypothesis. These findings justify using state-based models such as Markov Chains, which can better capture market dependencies.

1.3 Discretisation of raw Data

Discretisation is a fundamental step in simplifying continuous financial data for state-based modelling. In this analysis, we transform the daily price increments of the Nikkei 225 and FTSE 100 indices into a finite set of discrete states $\{-1, 0, +1\}$, defined as:

- $+1$: A positive price movement.
- 0: No significant change.
- -1: A negative price movement.

This transformation allows us to apply Markov Chain models, which require a discrete and finite state space.

Mesh Size Calculation

To ensure that the discretisation accurately reflects market behaviour without excessive information loss, we define a mesh size δ, which sets the threshold for classifying increments into states. This is calculated as:

$$\delta = 1.5 \times \text{Standard Deviation of Original Increments}$$

This choice ensures that most increments fall within the defined state space while preserving extreme fluctuations in a controlled manner.

Calculated Mesh Sizes

Table 2 presents the calculated mesh sizes for both indices. The FTSE 100 has a mesh size of $\delta = 66.08853$, while the Nikkei 225 has a mesh size of $\delta = 263.5396$.

Index	**Mesh Size**
FTSE 100	66.08853
Nikkei 225	263.5396

Table 2: Calculated Mesh Sizes for Discretisation

Initial Discretisation Results

Before adjustment, the initial discretisation resulted in the distributions shown in Table 3. This table illustrates the frequency of different state values before limiting extreme values.

States	**-3**	**-2**	**-1**	**0**	**1**	**2**	**4**
FTSE 100 Count	1	1	50	144	48	7	-
Nikkei 225 Count	1	1	33	162	42	4	1

Table 3: Initial Discretisation Results for FTSE 100 and Nikkei 225.

Adjustment of Extreme Values

A small number of observations ($< 5\%$) exceeded the ± 1 bounds, which could distort state-based modelling. To ensure consistency with the Markov model's state structure, any values outside the defined range were capped and reassigned. The adjusted distributions are shown in Table 4. This table illustrates the adjusted state counts for both the FTSE 100 and Nikkei 225 indices.

States	-1	0	1
FTSE 100 Count	52	144	55
Nikkei 225 Count	34	162	47

Table 4: Adjusted Discretisation Results for FTSE 100 and Nikkei 225.

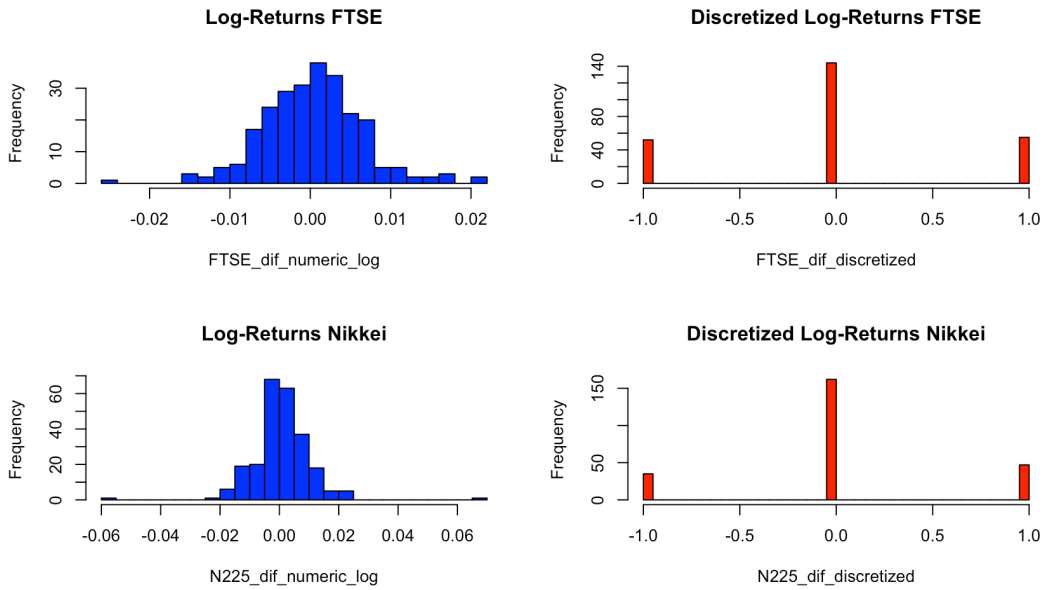


Figure 4

Impact of Discretisation

The discretisation process successfully simplifies complex financial data while preserving its essential characteristics. By reducing daily price changes to three representative states, we enable state-transition modelling without compromising interpretability or statistical integrity.

Although minor detail is lost due to collapsing extreme values into ± 1, this trade-off significantly enhances the tractability and robustness of the model. The discrete framework now allows us to fit a Markov Chain, which will be evaluated in the next section to determine if it captures underlying market dependencies more effectively than traditional models.

1.4 Markov Chain Fitting and Likelihood Ratio Test

To determine whether a Markov process effectively models stock price increments, we estimate the transition probabilities between states {-1,0,+1}, assuming that future price movements depend only on the current state. This contrasts with an independent increment model, where future changes are assumed to be unrelated to past movements. We used the `markovchain` **package** in R to construct a transition probability matrix, which represents the likelihood of moving from one state to another.

Estimated Transition Probability Matrix

The estimated transition probability matrix is shown below:

Current State	**-1**	**0**	**+1**
-1	0.35	0.35	0.30
0	0.19	0.60	0.21
+1	0.11	0.71	0.18

Table 5: Estimated Transition Probability Matrix for the Market States.

$$P = \begin{pmatrix} P(-1 \to -1) & P(-1 \to 0) & P(-1 \to +1) \\ P(0 \to -1) & P(0 \to 0) & P(0 \to +1) \\ P(+1 \to -1) & P(+1 \to 0) & P(+1 \to +1) \end{pmatrix} = \begin{pmatrix} 0.35 & 0.35 & 0.30 \\ 0.19 & 0.60 & 0.21 \\ 0.11 & 0.71 & 0.18 \end{pmatrix}$$

Where each element $P(i \to j)$ represents the probability of transitioning from state i to state j, with $i, j \in \{-1, 0, +1\}$.

If the market is currently in state 0 (no significant movement), there is a 60% probability it remains in state 0, a 19% probability of transitioning to -1 (negative movement), and a 21% probability of moving to +1 (positive movement).This matrix captures market dependencies, unlike independent models, which assume price changes are entirely random.

Independent Increment Model

For comparison, we construct an independent model where state transitions are determined solely by historical frequencies rather than conditional dependence. This model assumes that each price movement is independent of previous states.

To determine whether the Markov model outperforms the independent model, we use the Likelihood Ratio Test (LRT). The LRT compares the log-likelihood of the two models using the formula:

$$\lambda = -2\left(\mathcal{L}_{\text{independent}} - \mathcal{L}_{\text{Markov}}\right) \tag{1}$$

- $\mathcal{L}_{\text{independent}}$ is the log-likelihood of the independent model,

- $\mathcal{L}_{\text{Markov}}$ is the log-likelihood of the Markov model.

This test statistic follows a chi-square distribution with the degrees of freedom given by:

$$\text{df} = k^2 - k \tag{2}$$

where k is the number of states (in this case, $k = 3$, resulting in df $= 6$).

The log-likelihood curve $\mathcal{L}$ represents the likelihood of observing the data under different model assumptions, and the likelihood ratio test compares how well the two models (independent vs. Markov) fit the data.

Test Results

Metric	**Value**
Likelihood Ratio Test Statistic	16.20158
Degrees of Freedom	6
P-value	0.0127

Since the p-value (0.0127) < 0.05, we reject the null hypothesis, confirming that the Markov model fits the data significantly better than the independent model. This result implies that future price increments are dependent on current price movements, validating that stock prices exhibit Markovian properties rather than purely random fluctuations. The Markov model effectively captures short-term dependencies, making it a more appropriate tool for market forecasting and risk management compared to traditional independent increment models.

This insight underscores the importance of state-based modelling techniques, demonstrating that past price movements influence future market behaviour, rather than prices following an entirely random walk.

1.5 Confidence Interval Construction

In this analysis, we compare the forecasting performance of two models for price movements:

1. A random walk model with normally distributed increments (the original model).
2. A discretized Markov Chain model with state-dependent transitions.

Constructing Confidence Intervals Using Original Data

The random walk model assumes that price changes follow an independent normal distribution. Using the mean (μ) and standard deviation (σ) of daily increments, we simulate 10,000 future price paths. The 2.5% and 97.5% quantiles are extracted to form a 95% confidence interval.

Index	Forecasted Price	95% Confidence Interval
Nikkei 225	21,199.12	[17,960.45, 24,489.49]
FTSE 100	7,612.79	[6,800.62, 8,437.92]

Table 6: Random Walk Model Forecasting Results

Constructing Confidence Intervals Using the Discretized Markov Chain Model

The Markov model-based forecast follows a similar approach but simulates future states using Markov transition probabilities instead of assuming independent increments. The confidence interval is constructed using 10,000 simulated price paths, incorporating state-dependent behavior.

Index	Forecasted Price	95% Confidence Interval
Nikkei 225	21,189.99	[18,343.94, 23,878.27]
FTSE 100	7,456.63	[6,575.34, 8,293.64]

Table 7: Markov Chain Model Forecasting Results

Comparison of Results

By comparing the two approaches, we assess the effectiveness of discretization in forecasting market movements. The results indicate that both models provide similar forecasts, but the Markov model introduces a more realistic representation of state-dependent behavior, especially in the FTSE 100 case, where the confidence intervals are slightly narrower in comparison to the random walk model.

1.6 Factors Influencing the Fluctuations

The actual observed prices on 31 October 2017 were:

Index	Value
Nikkei 225	22,011.61
FTSE 100	6,694.00

Table 8: Observed values of the Nikkei 225 and FTSE 100 indices on 31 October 2017.

The Markov model performed better for FTSE 100, correctly capturing the actual price. Both models performed well for Nikkei 225, as the actual value fell within both confidence intervals.

Factors Influencing the Fluctuations:

- **Global Market Correlations:** Both the Nikkei 225 (Japan) and FTSE 100 (UK) exhibit strong correlations with global market trends. These trends are influenced by macroeconomic conditions such as interest rates, geopolitical risks, and central bank policies.

- **Deviations in FTSE 100:** The deviation of the FTSE 100 from the random walk model suggests that factors such as local economic performance, policy decisions, and market shocks may be contributing to its behavior.
- **Markov Model Advantages:** The Markov model accounts for dependency in market movements, improving forecasting accuracy for the FTSE 100, especially when the market experiences disruptions or when non-independence in the price movements is significant.

Conclusion

Markov models effectively capture dependencies in market movements, making them more robust in dynamic market conditions. Our analysis confirms that stock indices do not follow purely independent processes, highlighting the significance of state-based modeling techniques in accurately forecasting market behavior and capturing structural dependencies within financial data.

2 Question 2

2.1 Types of Censoring in Data

In our dataset, **right censoring** is present for both Type I and Type II, as some patients withdrew from the study, leaving us with no way to determine when they may have experienced a relapse. Patient withdrawals represent a form of **random censoring**, as they occurred at different time points throughout the study. Additionally, right censoring also occurs for patients who remained in remission until the end of the study, as their exact relapse time (if any) remains unknown. We do not observe left or interval censoring in this dataset. Based on this information, we conclude that right censoring, including random censoring due to patient withdrawals and administrative censoring due to the study's conclusion, is the only type of censoring present in the data.

2.2 Kaplan-Meier Estimate of the Time in Remission

2.2.1 Construct a Kaplan-Meier estimate

The Kaplan–Meier estimate at time t is given by:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right),$$

where:

- t_i are the ordered times at which relapses occur.
- d_i is the number of events at time t_i.
- n_i is the number of patients "at risk" just before time t_i.

The hazard function λ_j (often denoted as the hazard rate) can be calculated using the following formula:

$$\hat{\lambda_j} = \frac{d_j}{n_j}$$

Where:

- d_j is the number of events (e.g., deaths or failures) that occurred at time t_j,
- n_j is the number of individuals at risk just before time t_j.

Table 9: Kaplan-Meier Estimate for Type I Chemotherapy

J	t_j (Weeks)	d_j (Events)	c_j (Censored)	n_j (At Risk)	λ_j (Hazard)	$S(t)$ (Survival)
0	0	0	0	12	0.0000	1.0000
1	5	2	0	12	0.1667	0.8333
2	8	2	0	10	0.2000	0.6667
3	12	1	0	8	0.1250	0.5833
4	23	1	0	7	0.1429	0.5000
5	27	1	0	6	0.1667	0.4167
6	30	1	0	5	0.2000	0.3333
7	33	1	0	4	0.2500	0.2500
8	43	1	0	3	0.3333	0.1667
9	45	1	0	2	0.5000	0.0833

The Kaplan-Meier survival function $S(t)$ for type I is given by:

$$S(t) = \begin{cases} 1.00000, & 0 \leq t < 5 \\ 0.83333, & 5 \leq t < 8 \\ 0.66667, & 8 \leq t < 12 \\ 0.58333, & 12 \leq t < 23 \\ 0.50000, & 23 \leq t < 27 \\ 0.41667, & 27 \leq t < 30 \\ 0.33333, & 30 \leq t < 33 \\ 0.25000, & 33 \leq t < 43 \\ 0.16667, & 43 \leq t < 48 \\ 0.08333, & 48 \leq t < 161 \end{cases}$$

Table 10: Kaplan-Meier Estimate for Type II Chemotherapy

J	t_j (Weeks)	d_j (Events)	c_j (Censored)	n_j (At Risk)	λ_j (Hazard)	$S(t)$ (Survival)
0	0	0	0	11	0.000000	1.000000
1	9	1	0	11	0.090909	0.909091
2	13	1	1	10	0.100000	0.818182
3	18	1	0	8	0.125000	0.715909
4	23	1	0	7	0.142857	0.613636
5	31	1	0	6	0.166667	0.511364
6	34	1	0	5	0.200000	0.409091
7	48	1	0	4	0.250000	0.306818

The Kaplan-Meier survival function $S(t)$ is given by:

$$S(t) = \begin{cases} 1.00000, & 0 \leq t < 9 \\ 0.909091, & 9 \leq t < 13 \\ 0.818182, & 13 \leq t < 18 \\ 0.715909, & 18 \leq t < 23 \\ 0.613636, & 23 \leq t < 31 \\ 0.511364, & 31 \leq t < 34 \\ 0.409091, & 34 \leq t < 48 \\ 0.306818, & 48 \leq t < 161 \end{cases}$$

2.2.2 Graphical Representation of Survival and Hazard Function

Figure 5: Kaplan-Meier survival function $S(t)$ for Type I (standard chemotherapy)

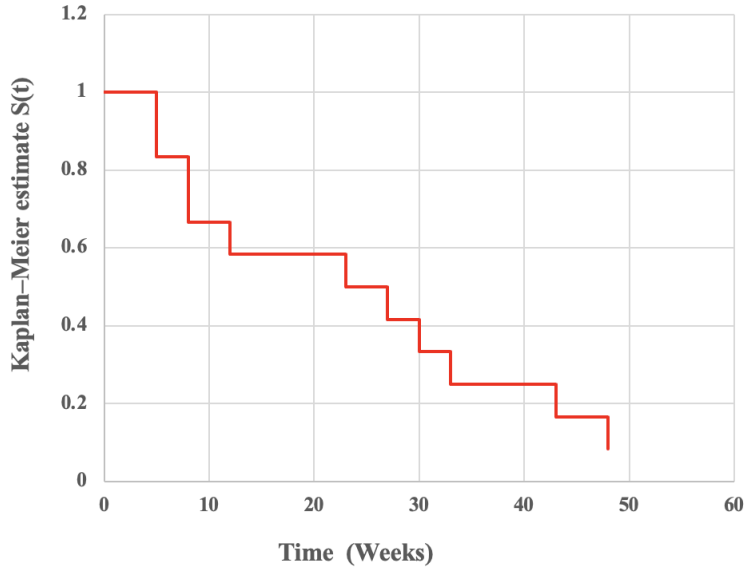


Figure 6: Hazard function $h(t)$ for Type I (standard chemotherapy)

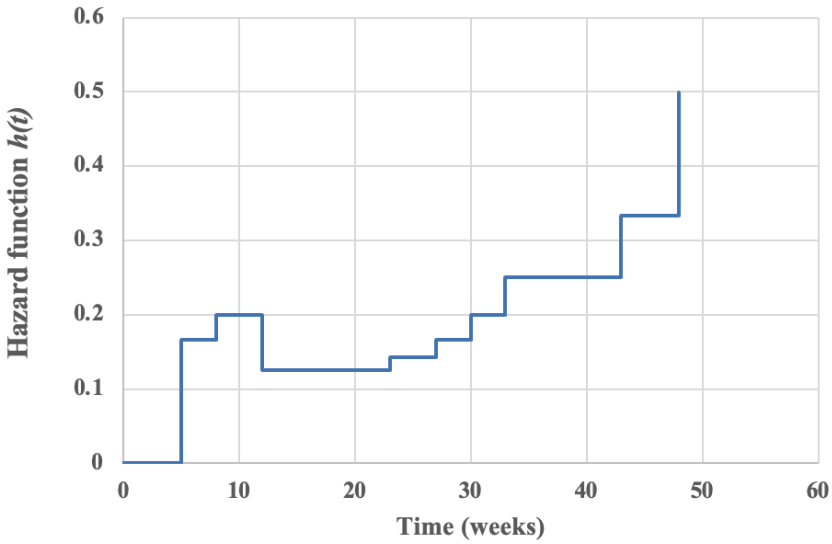


Figure 7: Kaplan-Meier survival function $S(t)$ for Type II (new chemotherapy)

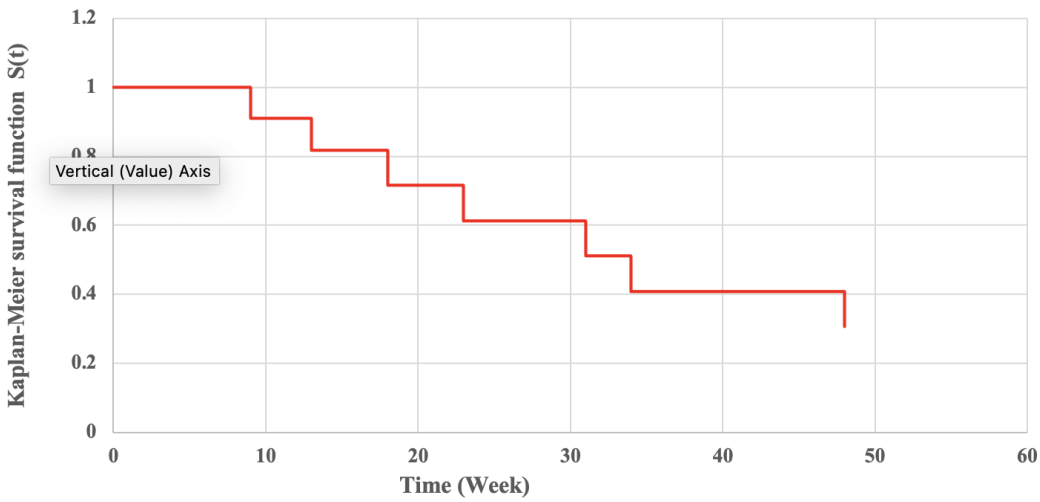
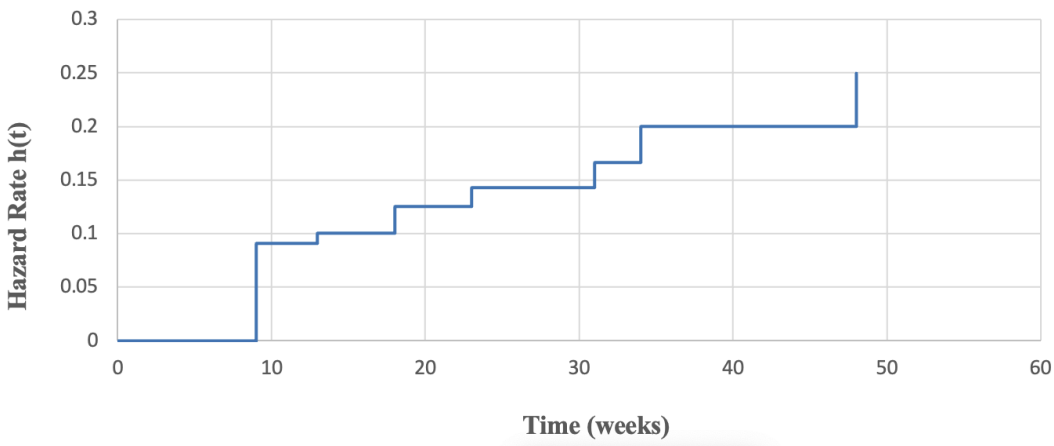


Figure 8: Hazard function $h(t)$ for Type II (new chemotherapy)



2.2.3 Conclusion on Whether the New Treatment Results in Greater Remission Time

The data and the Kaplan-Meier survival graph both indicate that Type II chemotherapy leads to a longer remission time compared to Type I chemotherapy. The graph clearly shows that patients treated with Type II chemotherapy remain in remission for a more extended period, with the survival rate for Type II consistently higher than that for Type I at various time intervals. This suggests that Type II chemotherapy is more effective at prolonging remission, as evidenced by both the data and the graphical representation. Therefore, the new treatment (Type II) demonstrates a clear advantage over the standard Type I chemotherapy in terms of delaying relapse and prolonging the remission period.

2.3 Kaplan-Meier Assumptions and Applicability

The Kaplan-Meier estimator relies on several key assumptions:

1. **Non-informative censoring:** This assumption implies that censored data (patients who withdrew) have the same probability of experiencing the relapse as those who are not censored. In this scenario, some patients withdrew from the study or were censored at different time points. If the reason for withdrawal is related to the treatment effectiveness (for example, patients experiencing fewer relapses on Type II treatment might be less likely to withdraw), this assumption could be violated.

2. **Independent survival times:** The survival times of patients are assumed to be independent of each other, meaning the survival time of one patient does not influence the survival time of another. In this case, the assumption seems reasonable, as the patients' remission times are independent of each other.

3. **Homogeneous lives:** The assumption here is that all patients are comparable in terms of their health status when they entered the study. It is assumed that no patients were in better or worse conditions when the trial began. In this case, since patients were randomly assigned to Type I or Type II treatments, this assumption is likely valid, although some variability in patient health might still exist.

4. **Homogeneous treatment effect:** The Kaplan-Meier estimator assumes that the treatment effect is the same for all patients. In this case, this assumption holds if both Type I and Type II treatments have the same impact across all patients. If there are variations in how patients respond to each treatment, this assumption could be violated.

2.4 Probability of Remission After 30 Weeks

To calculate the probability that a patient remains in remission after 30 weeks for **Type I** chemotherapy, we refer to the Kaplan-Meier survival estimate. From the survival table, we know the survival probability in the interval $30 < t < 33$ is:

$$P(30 < t < 33) = S(30 < t < 33) = 0.33333$$

Therefore, the probability that a patient remains in remission after 30 weeks for Type I chemotherapy is approximately:

$$P(\text{Remission at 30 weeks}) \approx 0.334$$

For **Type II** chemotherapy, the Kaplan-Meier estimate does not provide a direct survival probability at 30 weeks, as the dataset does not include a time point exactly at 30 weeks. However, we can approximate the survival probability at 30 weeks by combining the survival probabilities from the adjacent intervals.

The survival probabilities for the intervals are:

$$P(23 < t < 31) = S(23 < t < 31) = 0.613636$$

$$P(31 < t < 34) = S(31 < t < 34) = 0.511364$$

Given that 30 weeks falls within the interval $23 < t < 31$, we approximate the survival probability at 30 weeks using the value for this interval:

$$P(30 < t < 31) = S(23 < t < 31) = 0.613636$$

Thus, the probability that a patient remains in remission after 30 weeks for Type II chemotherapy is approximately:

$$P(\text{Remission at 30 weeks}) \approx 0.613636$$

For a more detailed overview, the table below summarizes the survival probabilities at 30 weeks for both Type I and Type II chemotherapy treatments:

Chemotherapy Type	**Interval**	**Survival Probability at 30 Weeks**
Type I	$30 < t < 33$	0.333
Type II	$30 < t < 31$	0.614

These results suggest that patients undergoing **Type II chemotherapy** have a higher probability of remaining in remission after 30 weeks compared to those undergoing **Type I chemotherapy**.

2.5 Hypothesis Test for Remission Probability

The manufacturer claims that 40% of patients will remain in remission after 30 weeks of treatment. To test this claim, we will conduct a one-sample proportion test at a significance level of $\alpha = 0.05$, comparing the observed survival probability with the manufacturer's claimed probability of 40%.

We define the following hypotheses:

- **Null Hypothesis (H_0)**: $p = 0.40$
  The proportion of patients in remission after 30 weeks is 40%.
- **Alternative Hypothesis (H_1)**: $p \neq 0.40$
  The proportion of patients in remission after 30 weeks is not equal to 40%.

Where, p is the true proportion of patients in remission after 30 weeks and $p_0 = 0.40$ is the manufacturer's claimed proportion.

We will use a Z-test for proportions to test the hypothesis. The test statistic is calculated using the formula:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Where, $\hat{p}$ is the sample proportion (observed survival probability), p_0 is the claimed proportion (0.40) and n is the sample size.

Test Calculation for Type I Chemotherapy

For Type I chemotherapy, the observed survival probability at 30 weeks is $\hat{p} = 0.34$, and the sample size is $n = 11$. Substituting these values into the formula:

$$Z = \frac{0.34 - 0.40}{\sqrt{\frac{0.40(1-0.40)}{11}}} = -0.406$$

The corresponding p-value is:

$$\text{P-value} = 0.689$$

Test Calculation for Type II Chemotherapy

For Type II chemotherapy, the observed survival probability at 30 weeks is $\hat{p} = 0.614$, and the sample size is $n = 12$. Substituting these values into the formula:

$$Z = \frac{0.614 - 0.40}{\sqrt{\frac{0.40(1-0.40)}{12}}} = 1.510$$

The corresponding p-value is:

$$\text{P-value} = 0.13104$$

The p-values for both **Type I chemotherapy** (0.689) and **Type II chemotherapy** (0.13104) are greater than the significance level of 0.05, leading us to fail to reject the null hypothesis in both cases. Therefore, there is insufficient evidence to support the manufacturer's claim that 40% of patients will remain in remission after 30 weeks of treatment. The observed survival probabilities—33.4% for Type I and 61.4% for Type II—are both inconsistent with the 40% claim.

Given the lack of evidence supporting the manufacturer's claim, it is recommended that they refrain from stating that 40% of patients will be in remission after 30 weeks. Further investigation, possibly with a larger sample size or refined analysis, would be needed to determine a more accurate estimate of remission rates at 30 weeks.

3 Question 3

3.1 Probability of Functioning After 60 Hours

Determine the probability that a new part continues to function properly after 60 hours of testing.

Given the hazard function

$$\mu(x) = 0.00018 \cdot 1.06^x,$$

To find the probability that a part functions properly after 60 hours, we need to calculate the survival function:

$$S(x) = \exp\left(-\int_0^x \mu(t)\,dt\right)$$

We will first calculate the cumulative hazard function by integrating $\mu(x)$ from 0 to 60 hours:

$$\int_0^t \mu(x)\,dx = \int_0^{60} 0.00018 \cdot (1.06^x)\,dx$$

This is an exponential function in the form of $\int 1.06^x\,dx$, which can be solved by the following integration rule:

$$\int a^x\,dx = \frac{a^x}{\ln(a)} + C$$

Substituting $a = 1.06$, we get:

$$\int 0.00018 \cdot 1.06^x\,dx = 0.00018 \cdot \frac{1.06^x}{\ln(1.06)} \tag{3}$$

Now, we substitute the limits of integration from 0 to 60:

$$\int_0^{60} 0.00018 \cdot (1.06^x)\,dx = 0.00018 \cdot \left(\frac{1.06^{60}}{\ln(1.06)} - \frac{1.06^0}{\ln(1.06)}\right)$$

Since $1.06^0 = 1$, this simplifies to:

$$= 0.00018 \cdot \left(\frac{1.06^{60}}{\ln(1.06)} - \frac{1}{\ln(1.06)}\right)$$

Finally, the survival probability is given by

$$S(60) = \exp\left(-0.00018 \cdot \left(\frac{1.06^{60} - 1}{\ln(1.06)}\right)\right) = \exp(-0.098814)$$

$$S(60) = 0.90591$$

3.2 Conditional Survival Probability

The probability that it continues to function properly after 50 hours is given by the conditional probability:

$$P(T > 50 \mid T > 20) = \frac{S(50)}{S(20)}$$

Where $S(x)$ is the survival function. The solution for the survival function is given in equation (1) .

For $t = 20$: from equation 3 we have:

$$S(20) = \exp\left(-0.00018 \cdot \left(\frac{1.06^{20} - 1}{\ln(1.06)}\right)\right)$$

$$S(20) = 0.993205071$$

For $t = 50$: from equation 3 we have:

$$S(50) = \exp\left(-0.00018 \cdot \left(\frac{1.06^{50} - 1}{\ln(1.06)}\right)\right)$$

$$S(50) = 0.9476092438$$

Finally, we calculate the conditional probability:

$$P(T > 50 \mid T > 20) = \frac{S(50)}{S(20)} = \frac{0.9476092438}{0.993205071} = 0.9541$$

Thus, the probability that the part continues to function properly after 50 hours, given that it is functioning properly after 20 hours, is approximately 0.9541.

3.3 Median Failure Time Calculation

To find the time t when 50% of new parts are expected to malfunction, we set the survival function $S(t) = 0.5$: We have equation (1):

$$S(t) = \exp\left(-0.00018 \times \frac{1.06^t - 1}{\ln(1.06)}\right)$$

Setting $S(t) = 0.5$, we get:

$$\exp\left(-0.00018 \times \frac{1.06^t - 1}{\ln(1.06)}\right) = 0.5$$

Taking the natural logarithm of both sides:

$$-0.00018 \times \frac{1.06^t - 1}{\ln(1.06)} = \ln(0.5)$$

Solving for t:

$$\frac{1.06^t - 1}{\ln(1.06)} = \frac{\ln(0.5)}{-0.00018}$$

$$\frac{1.06^t - 1}{0.0582689} = \frac{0.693147}{0.00018} = 3850.81767$$

$$1.06^t - 1 = 224.3829097$$
$$1.06^t = 225.3829097$$

Taking the natural logarithm of both sides:

$$t \ln(1.06) = ln(225.3829097) = 5.4178$$

Finally, solving for t:

$$t = \frac{5.417800}{\ln(1.06)}$$

After performing the calculation:

$$t \approx 92.98 \,\text{hours}$$

Thus, the median malfunction time, when 50% of parts are expected to malfunction, is approximately **92.98 hours**.

3.4 Alternative Hazard Function Formulation

We propose a proportional hazards model where the hazard function is adjusted for real-world factors, incorporating both driving style and road quality as covariates. Let the baseline hazard function be denoted as $\mu(x)$, where x represents the baseline covariates. The effect of the covariates is modeled through a multiplicative function $g(z)$, where z is the vector of covariates corresponding to driving style and road quality.

The driving style covariate D is treated as a categorical variable with three levels:

$$D \in \{0, 1, 2\}, \quad \text{where} \quad \begin{array}{ll} D = 0 & \text{represents "careful driving",} \\ D = 1 & \text{represents "normal driving",} \\ D = 2 & \text{represents "reckless driving".} \end{array}$$

The road quality covariate R is modeled as an indicator variable:

$$R \in \{0, 1\}, \quad \text{where} \quad \begin{array}{ll} R = 0 & \text{corresponds to urban areas (better road quality),} \\ R = 1 & \text{corresponds to rural areas (poorer road quality).} \end{array}$$

3.5 Baseline Hazard Function

A sample of drivers are asked to record the length of time the part was used before it malfunctioned. The following Cox regression model was used to fit the data: A Cox regression model was used to fit the data as detailed below:

$$h_i(t) = h_0(t) \exp\left(-0.12(x_i - 3) + 0.3g_i - 0.2c_i\right)$$

where:

$$x_i = \text{average time of individual session where the car is driven,}$$
$$g_i = \begin{cases} 1 & \text{if the car was driven in both the city and rural areas,} \\ 0 & \text{if the car was driven only in the city,} \end{cases}$$

$$c_i = \begin{cases} 0 & \text{if the car was serviced at the designated time intervals,} \\ 1 & \text{if the car was not serviced at the designated time intervals.} \end{cases}$$

To obtain the baseline hazard, we need to set the terms in the exponential part that result in zero. This occurs when:

- The average time of individual driving sessions, denoted as x_i, is 3 hours.
- The car is driven exclusively in the city, meaning $g_i = 0$.
- The car is serviced at regular intervals, represented by $c_i = 0$.

Thus, the baseline hazard function reflects the hazard rate under these conditions: an average driving session time of 3 hours, city-only driving, and regular servicing at designated intervals. It is represented as $h_0(t)$.

Mathematically,

$$h_0(t) = h_i(t)\Big|_{x_i=3,\, g_i=0,\, c_i=0} = h_0(t)\exp(-0.12 \cdot (3-3) + 0.3 \cdot 0 - 0.2 \cdot 0)$$

Since the terms involving g_i and c_i are zero, the expression simplifies to:

$$h_0(t) = h_0(t)\exp(0) = h_0(t)$$

Therefore, the baseline hazard function is $h_0(t)$, where the terms corresponding to x_i, g_i, and c_i reflect the specific conditions outlined above.

3.6 Survival Function Comparison

Case 1: Car driven only in the city i.e.$g_i = 0$, $x_i = 2$ hours, not maintained i.e. $c_i = 1$. The hazard function $h_1(t)$ is given by:

$$h_1(t) = h_0(t)\text{exp}(-0.12 \times (2-3) + 0.3 \times 0 - 0.2 \times 1)$$

$$h_1(t) = h_0(t)\text{exp}(0.12 - 0.2)$$

$$h_1(t) = h_0(t)\text{exp}(-0.08)$$

Case 2: Car driven in both city and rural areas $g_i = 1$, $x_i = 4$ hours, not maintained $c_i = 1$.The hazard function $h_2(t)$ is given by:

$$\begin{aligned} h_2(t) &= h_0(t)\text{exp}(-0.12 \times (4-3) + 0.3 \times 1 - 0.2 \times 1) \\ &= h_0(t)\text{exp}(0.12 + 0.3 - 0.2) \\ &= h_0(t)\text{exp}(-0.02) \end{aligned}$$

Taking the ratio of the two hazard functions:

$$\frac{h_1(t)}{h_2(t)} = \frac{h_0(t)\exp(-0.08)}{h_0(t)\exp(-0.02)}$$

Since $h_0(t)$ appears in both the numerator and denominator, it cancels out:

$$\frac{h_1(t)}{h_2(t)} = \frac{\exp(-0.08)}{\exp(-0.02)}$$

Using the property of exponentials $\frac{\exp(a)}{\exp(b)} = \exp(a-b)$:

$$\frac{h_1(t)}{h_2(t)} = \exp(0.02 - 0.08) = \exp(-0.06)$$

Thus, the ratio of the hazard functions is:

$$\frac{h_1(t)}{h_2(t)} = 0.94176 \tag{4}$$

Now, we can find the survival function for case 1:

$$S_1(t) = \exp\left(-\int_0^t h_1(t)\,dt\right)$$

form equation (4) above, we have

$$h_1(t) = 0.94176 \times h_2(t)$$

substituting we get

$$S_1(t) = \exp\left(-\int_0^t 0.94176 \times h_2(t)\,dt\right)$$

which is

$$S_1(t) = S_2(t)^{0.94176}$$

clearly,

$$S_1(t) > S_2(t)$$

The survival function for the part where the car is driven only in the city for a shorter duration is higher than the survival function for the part where the car is driven in both city and rural areas for a longer duration, with the car not being maintained.

3.7 Impact of Driving Style on Model Accuracy

A high standard error in the model may stem from several factors. One common cause is a small sample size, which can lead to imprecise estimates and reduce the reliability of the results. Another potential issue is measurement error, such as inaccurate recording of driving sessions, which can introduce noise and distort the findings. Additionally, the model may not account for all relevant factors influencing part failure, leading to omitted variable bias and affecting the precision of the estimates.

Including driving style as a covariate in the model could enhance its accuracy by capturing the impact of specific driving behaviors, such as harsh braking, aggressive maneuvers, and other driving habits that contribute to part wear and tear. Moreover, driving style could account for other factors, such as gender and occupation, which may also influence driving patterns and their effect on part durability.

However, there are challenges in defining and measuring driving style. Since driving style is inherently subjective, it can be prone to bias, making it difficult to standardize and quantify consistently. Additionally, collecting data on driving styles can be costly and logistically challenging. Another issue is the potential correlation between driving style and other variables, such as car maintenance practices, which could lead to multicollinearity and distort the model's estimates.

3.8 Additional Covariates for Model Improvement

Two additional covariates that could improve the model's predictability are **Car Maintenance History** and **Environmental Conditions** (e.g., Climate).

1. Car Maintenance History

Reason for inclusion:

- Regular maintenance (e.g., oil changes, tire replacements, brake checks) significantly influences the longevity and performance of car parts. By including maintenance history as a covariate, the model can account for how well the car is maintained, which affects part wear and tear.
- Cars with poor maintenance are likely to experience failures more quickly, and including this covariate can improve the accuracy of the model by reflecting this relationship.

Problems that may arise:

- **Data Availability:** Maintenance records might be incomplete or inconsistent, making it difficult to quantify this covariate accurately.
- **Self-Reported Data:** If maintenance is self-reported (e.g., by the owner), it may be unreliable or biased, leading to measurement errors.
- **Complexity of Maintenance Factors:** The specific type of maintenance and the frequency of repairs might vary significantly, making it challenging to model the full effect properly.

2. Environmental Conditions (e.g., Climate)

Reason for inclusion:

- Environmental factors such as temperature extremes and humidity play a major role in the wear and tear of car parts. Including these factors as covariates could help the model account for these external influences, improving its predictive power.
- For example, cars in areas with harsh climates may experience more frequent breakdowns or faster deterioration of parts.

Problems that may arise:

- **Data Collection:** Environmental data can be difficult to gather accurately, especially for each individual car. Road conditions may vary locally, and climate data might not be easily available on a granular level.

- **Complex Interactions:** The impact of environmental conditions could interact with other covariates (e.g., driving style), complicating the interpretation and potentially leading to multicollinearity or overfitting if not modeled carefully.